3. GLUT2 on the basolateral membrane, by facilitated diffusion, transports the glucose from the enterocytes into the blood circulation.

- Glucose and galactose are absorbed by the enterocyte by cotransport with sodium using the same transporter.
- Thus, the sodium dependent SGLT1 and sodium independent GLUT2 play major role in absorption of glucose.

Carbohydrates that are not digested and absorbed in the digestive tract so far will reach the large intestine where they are broken down by intestinal bacteria. Indigestible fibers and other carbohydrates are excreted with feces.

Factors affecting absorption

- Absorption is high with intact mucosa. Any damage to the intestinal mucosa in the form of infections and congenital disorders reduce the rate of absorption.
- Hormones such as thyroid hormones, pituitary hormones and mineralocorticoids increase the absorption, while insulin has no effect on absorption of carbohydrates.
- Vitamin B6, B12 and pantothenic acid deficiencies decrease the absorption of monosaccharides.
- Advancing age decreases the absorption.
- Inhibitors of carbohydrate digestive enzymes and SGLT1 inhibitors like phlorizin decrease the absorption.

2.2.2.2 Digestion and absorption of proteins

Proteins have to be broken down to it's constituent amino acids for absorption into the gut. The digestion and absorption of proteins is very effective in a healthy human. The major contribution of the protein is from diets rich in protein like milk, meat, egg, pulses, soybean, nuts, etc. and minor contribution is from the proteins present in the digestive juices and worn out intestinal epithelial cells that are shed into the lumen.

Digestion in mouth

Proteolytic enzymes are not present in saliva and therefore, no digestion of proteins takes place in the mouth. However, chewing the protein-rich foods especially the harder ones like meat increases the surface area of the food particles and thereby allows digestion to occur more quickly.

Digestion in stomach

In the stomach the proteins are converted into: Protein → Metaprotein → Proteone → Peptone → Peptide

HCl: The partially denatured proteins (denaturation occurring due to cooking process) or the non-denatured proteins present in the raw foods are further denatured by the highly acidic environment present in the stomach, which unveils the sites for action of the peptidases. HCl also initiates the conversion of inactive pepsinogen to active enzyme pepsin.

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